HW2：線上資料擷取

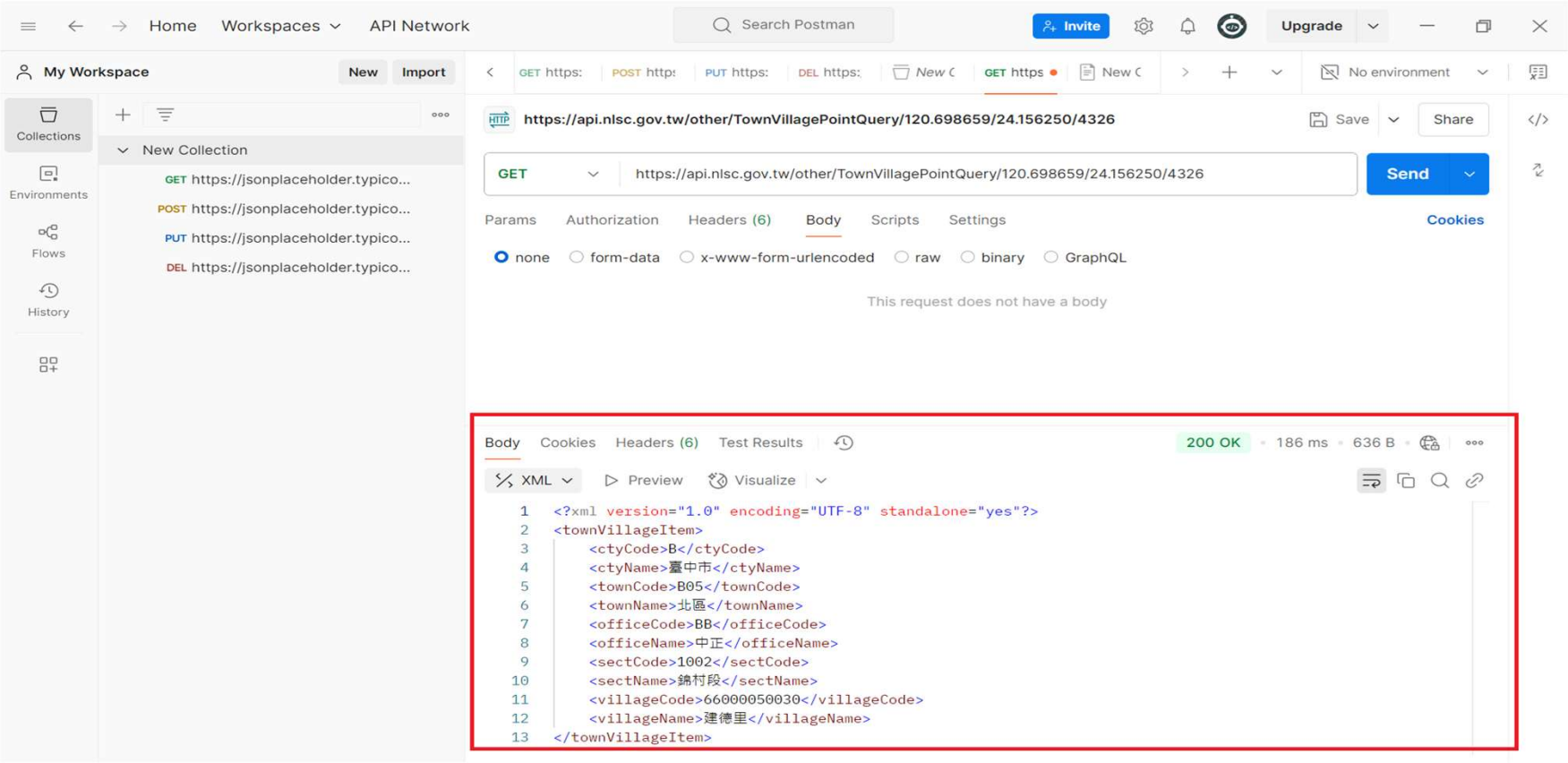
HW2 Requirements

- 自行挑選一個提供API串接的線上平台，撰寫程式透過API抓取資料，並儲存成JSON或CSV格式 (不能跟課堂講義範例相同)
- 建立一GitHub Actions Workflow，執行上述程式
  - 每天觸發一次及手動觸發該Workflow

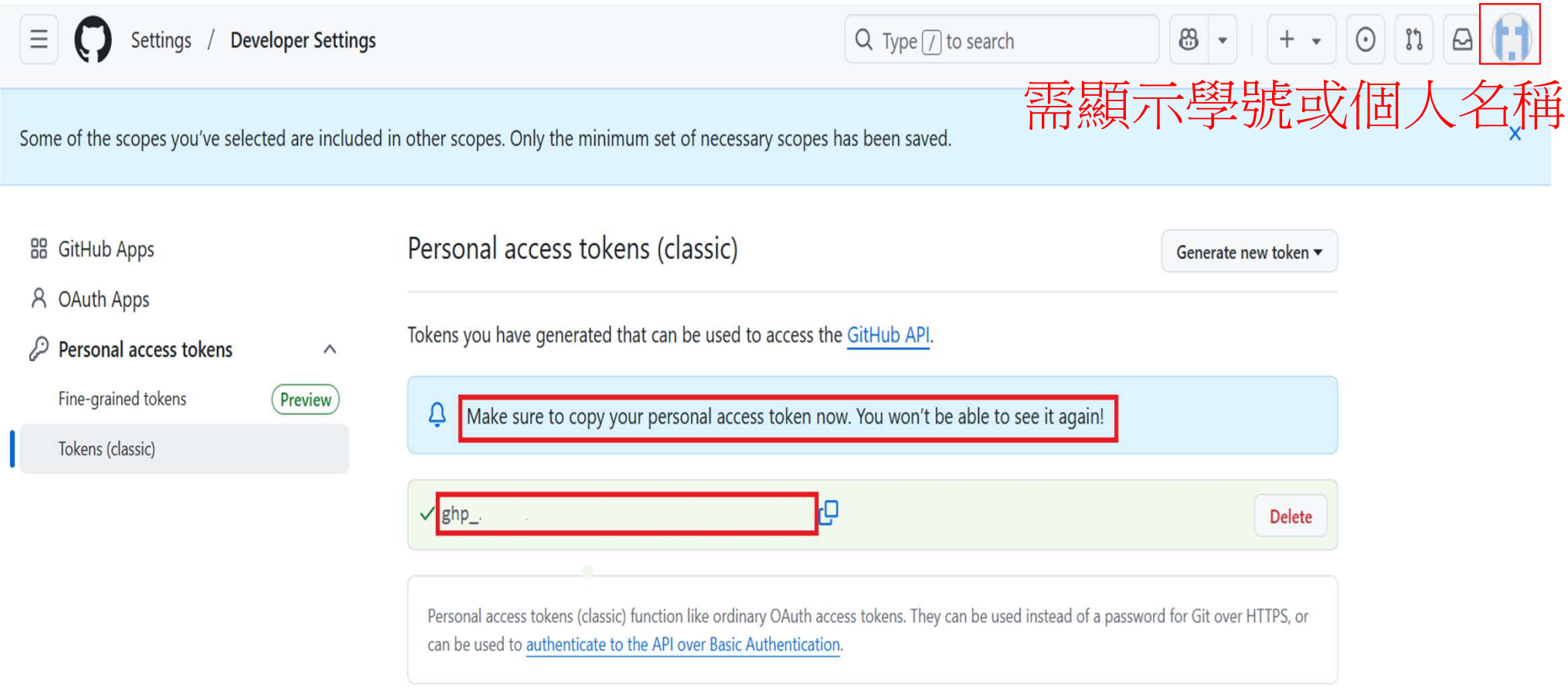
HW2 Requirements

- (10%) Postman模擬HTTP GET或HTTP POST請求結果截圖 (並說明)
  - 無說明扣5%
- (10%) GitHub Personal Access Token截圖
  - 無顯示個人名稱示扣5%
- (30%) GitHub Actions Workflow每天觸發一次及手動觸發截圖
  - 無顯示個人名稱示扣15%
- (40%) 程式碼與YAML檔 (並說明)
  - 無貼上程式碼與YAML檔扣20%、無說明扣20%
- (10%) 輸出JSON或CSV資料 (並說明)
  - 無貼上輸出JSON或CSV資料扣5%、無說明扣5%
- 書面報告 (檔名：111xxxx.pdf)，內容包含以上五項

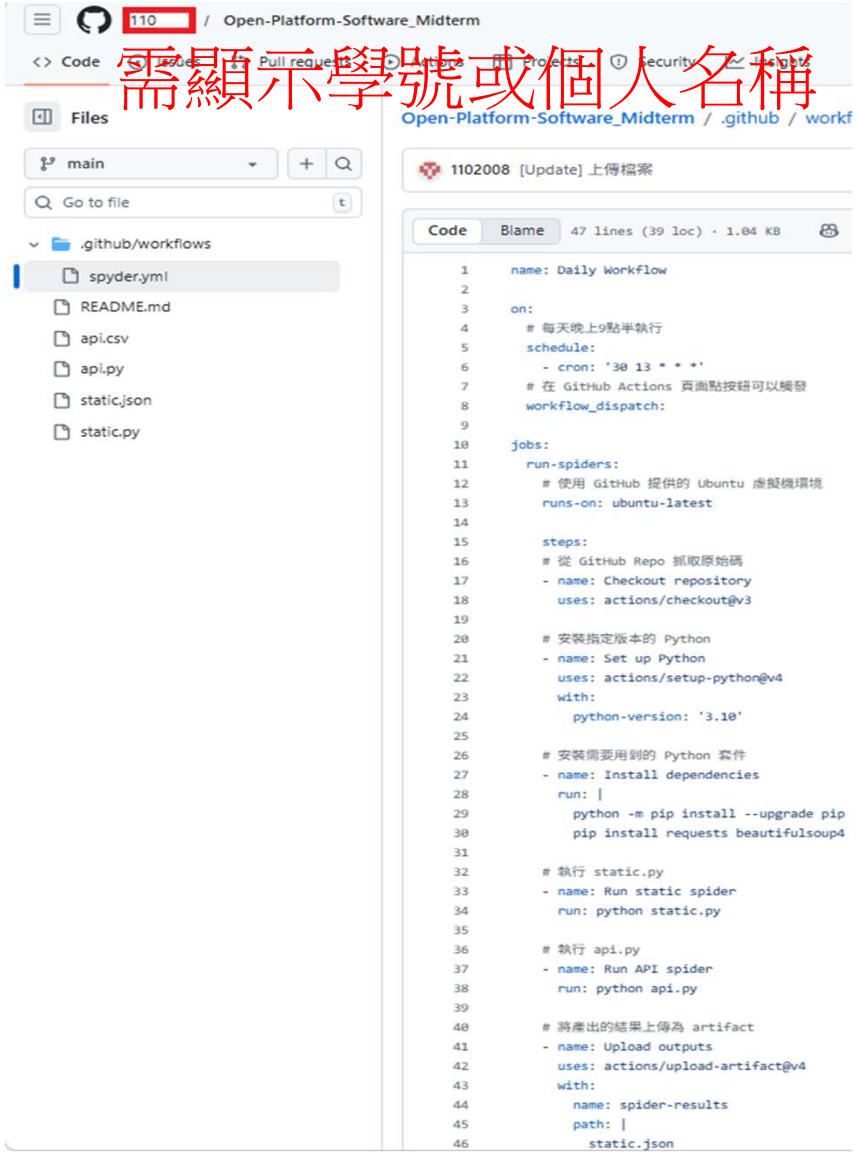
HW2 參考資訊


Home
Workspaces
API Network
Search Postman
Invite
Upgrade
My Workspace
New
Import
Collections
Environments
Flows
History
New Collection
GET https://jsonplaceholder.typico...
POST https://jsonplaceholder.typico...
PUT https://jsonplaceholder.typico...
DEL https://jsonplaceholder.typico...
No environment
https://api.nlsc.gov.tw/other/TownVillagePointQuery/120.698659/24.156250/4326
Save
Share
GET
https://api.nlsc.gov.tw/other/TownVillagePointQuery/120.698659/24.156250/4326
Send
Params
Authorization
Headers (6)
Body
Scripts
Settings
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
GraphQL
This request does not have a body
Body
Cookies
Headers (6)
Test Results
200 OK
186 ms
636 B
XML
Preview
Visualize
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<townVillageItem>
<ctyCode>B</ctyCode>
<ctyName>臺中市</ctyName>
<townCode>B05</townCode>
<townName>北區</townName>
<officeCode>BB</officeCode>
<officeName>中正</officeName>
<sectCode>1002</sectCode>
<sectName>錦村段</sectName>
<villageCode>66000050030</villageCode>
<villageName>建德里</villageName>
</townVillageItem>

HW2 參考資訊


Settings / Developer Settings
Type / to search
需顯示學號或個人名稱
Some of the scopes you've selected are included in other scopes. Only the minimum set of necessary scopes has been saved.
GitHub Apps
OAuth Apps
Personal access tokens
Fine-grained tokens
Preview
Tokens (classic)
Personal access tokens (classic)
Generate new token
Tokens you have generated that can be used to access the GitHub API.
Make sure to copy your personal access token now. You won't be able to see it again!
ghp_
Delete
Personal access tokens (classic) function like ordinary OAuth access tokens. They can be used instead of a password for Git over HTTPS, or can be used to authenticate to the API over Basic Authentication.

HW2 參考資訊


110 / Open-Platform-Software_Midterm
Code
Pull requests
Actions
Projects
Security
Insights
需顯示學號或個人名稱
Files
main
Go to file
.github/workflows
spyder.yml
README.md
api.csv
api.py
static.json
static.py
Open-Platform-Software_Midterm / .github / workf
1102008 [Update] 上傳檔案
Code
Blame
47 lines (39 loc) · 1.04 KB
name: Daily Workflow
on:
每天晚上9點半執行
schedule:
- cron: '30 13 * * *'
在 GitHub Actions 頁面點按鈕可以觸發
workflow_dispatch:
jobs:
run-spiders:
使用 GitHub 提供的 Ubuntu 虛擬機環境
runs-on: ubuntu-latest
steps:
從 GitHub Repo 抓取原始碼
- name: Checkout repository
uses: actions/checkout@v3
安裝指定版本的 Python
- name: Set up Python
uses: actions/setup-python@v4
with:
python-version: '3.10'
安裝需要用到的 Python 套件
- name: Install dependencies
run: |
python -m pip install --upgrade pip
pip install requests beautifulsoup4
執行 static.py
- name: Run static spider
run: python static.py
執行 api.py
- name: Run API spider
run: python api.py
將產出的結果上傳為 artifact
- name: Upload outputs
uses: actions/upload-artifact@v4
with:
name: spider-results
path: |
static.json

HW2 Deadline

- 4/6(一)23:59前上傳至Portal
- 遲交1天分數9折、遲交2天分數8折、遲交3天分數7折、遲交4天分數6折，之後再補交者不予計分

References

- https://ithelp.ithome.com.tw/articles/10186119
- https://kucw.io/blog/github-actions-intro/